

Structured Source Coding: Shannon Theory as the Vanishing-Clustering Limit

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Abstract—Classical source coding treats the atypical set as a monolithic error event, achieving reliable compression at rates approaching the source entropy $H(X)$. We develop a *structured* extension in which the atypical set $B_\epsilon^{(n)}$ is partitioned into k clusters under a Hamming-distance metric. This induces a tunable parameter

$$\chi = \frac{\log_2 k}{n(H(X) + \epsilon)}$$

that controls clustering granularity. The effective rate becomes $R = (1 + \chi)(H(X) + \epsilon)$.

We prove that when the atypical set is clustered according to a distortion metric, the dominant-cluster probability satisfies

$$D \triangleq \Pr(X^n \in C_{\max}) \leq \min(\epsilon, 2^{2n\epsilon}/k),$$

with strict improvement $D < \epsilon$ in a nonempty finite-blocklength regime. Classical Shannon source coding emerges exactly as the limiting case $\chi \rightarrow 0$.

Exhaustive Monte-Carlo simulations on a Bernoulli($p = 0.3$) source ($H(X) \approx 0.881$ bits) demonstrate that *real Hamming-distance k -means clustering* exploits the geometric structure of the atypical set, yielding measurable reductions in D (e.g., $D = 0.0367 < 0.1$ for $n = 30, k = 16$). The structure in the atypical set is real, measurable, and exploitable. The refinement vanishes asymptotically, preserving consistency with Shannon's theorem.

Index Terms—Source coding, typical set, clustering, finite blocklength, error exponent, rate–error tradeoff.

I. INTRODUCTION

Shannon's source coding theorem guarantees that for any $\epsilon > 0$, the atypical set $B_\epsilon^{(n)}$ satisfies $\Pr(X^n \in B_\epsilon^{(n)}) \leq \epsilon$ for sufficiently large n . All sequences in $B_\epsilon^{(n)}$ are mapped to a single error symbol.

We introduce *structure* by partitioning $B_\epsilon^{(n)}$ into $k \geq 2$ clusters $\{C_1, \dots, C_k\}$ using a distortion metric (Hamming distance). The encoder now outputs:

- 1 indicator bit (typical vs. atypical),
- $n(H(X) + \epsilon)$ bits for typical sequences,
- $\log_2 k$ bits for the cluster label.

The clustering overhead χ yields the effective rate $R = (1 + \chi)(H(X) + \epsilon)$. When $\chi \rightarrow 0$, the scheme reduces exactly to classical source coding. For $\chi > 0$, we obtain a refined rate–error tradeoff that exploits the internal geometry of the atypical set.

II. CLASSICAL FRAMEWORK

Let $A_\epsilon^{(n)} = \{x^n : |-\frac{1}{n} \log_2 P(x^n) - H(X)| \leq \epsilon\}$ be the typical set and $B_\epsilon^{(n)}$ its complement. Then

$$\Pr(X^n \in B_\epsilon^{(n)}) \leq \epsilon, \quad |B_\epsilon^{(n)}| \leq 2^{n(H(X) + \epsilon)}.$$

Every $x^n \in B_\epsilon^{(n)}$ satisfies $P(x^n) \leq 2^{-n(H(X) - \epsilon)}$.

III. STRUCTURED CLUSTERING AND DOMINANT-CLUSTER BOUND

Theorem 1 (Refined Dominant-Error Bound via Structured Clustering (Upper Bound)). *Let X^n be an i.i.d. sequence from a discrete memoryless source with entropy $H(X)$. Partition the atypical set $B_\epsilon^{(n)}$ into $k \geq 2$ clusters $\{C_1, \dots, C_k\}$ according to any distortion metric. Let $C_{\max}$ be the cluster of largest cardinality. Then*

$$D \triangleq \Pr(X^n \in C_{\max}) \leq \min(\epsilon, 2^{2n\epsilon}/k).$$

Moreover, when the atypical set possesses geometric structure under the chosen metric, Hamming-distance k -means clustering empirically achieves a dominant probability D that is strictly smaller than ϵ in the finite-blocklength regime

$$n < \frac{1}{2\epsilon} \log_2(k/\epsilon).$$

Proof. In the worst-case equal-size partition, $c_{\max} \leq |B_\epsilon^{(n)}|/k \leq 2^{n(H(X) + \epsilon)}/k$. Since each sequence in $B_\epsilon^{(n)}$ has probability at most $2^{-n(H(X) - \epsilon)}$, we obtain

$$D \leq c_{\max} \cdot 2^{-n(H(X) - \epsilon)} \leq 2^{2n\epsilon}/k.$$

Combining with the classical bound $\Pr(B_\epsilon^{(n)}) \leq \epsilon$ yields the stated result. The strict improvement $D < \epsilon$ holds precisely when $k > 2^{2n\epsilon}/\epsilon$. $\square$

Lemma 1 (Matching Lower Bound – Worst-Case Tightness). *For any clustering into k parts, there exist sources (e.g., those for which the atypical mass is nearly uniformly distributed across the support of $B_\epsilon^{(n)}$) such that*

$$D \geq (1 - o(1)) \cdot \frac{2^{2n\epsilon}}{k}.$$

Thus the upper bound of Theorem 1 is tight in the worst case: no clustering scheme can achieve a better dominant-cluster probability than the combinatorial bound when the atypical set has uniform mass distribution.

Sketch. If the probability mass of $B_\epsilon^{(n)}$ is distributed approximately uniformly over its $|B_\epsilon^{(n)}|$ elements (possible when atypical types have comparable rate-function values), then any partition into k clusters satisfies $\max_i |C_i| \geq |B_\epsilon^{(n)}|/k$. Multiplying by the per-sequence probability bound $2^{-n(H(X)-\epsilon)}$ yields the stated lower bound on D . $\square$

The strict improvement $D \ll \epsilon$ therefore occurs *only* when the source geometry induces *cluster concentration*: the atypical mass is heavily concentrated in a small number of Hamming-balls (or type classes) near the typical-set boundary, as predicted by Sanov's theorem for convex rate functions.

Theorem 2 (Unequal Cluster Sizes). *Let the clusters have arbitrary cardinalities $c_1 \geq c_2 \geq \dots \geq c_k$. Then*

$$D \leq \min \left(\epsilon, \max_i c_i \cdot 2^{-n(H(X)-\epsilon)} \right).$$

When the partition is produced by Hamming-distance k -means, the largest cluster $C_{\max}$ is typically the one nearest the typical-set boundary, yielding the empirical scaling $D \approx \epsilon^{1+\chi}$.

The geometric-structure claim (concentration near the boundary) is verified empirically in Section ??.

a) *Structural refinement (empirical law).*: While the cardinality bounds provide worst-case limits, empirical evidence indicates that real sources exhibit strong non-uniformity within the atypical set under natural metrics. In particular, for clustering induced by Hamming distance on i.i.d. Bernoulli sources, the dominant-cluster probability empirically satisfies

$$D \approx \epsilon^{1+\chi},$$

suggesting that the atypical set decomposes into highly unequal components. This behavior is not implied by the combinatorial bounds and reflects underlying geometric structure in the source distribution.

b) *Conditional improvement.*: If the atypical set exhibits cluster concentration under the chosen metric, i.e., if there exists a partition such that

$$\max_i \Pr(X^n \in C_i) \gg \frac{\epsilon}{k},$$

then clustering yields a strictly stronger bound on the dominant error probability than uniform partitioning, and the empirical scaling $D \approx \epsilon^{1+\chi}$ becomes achievable.

Rate-error tradeoff. In regimes where the exponential term dominates, the empirical law above produces the explicit tradeoff curve

$$-\log_2 D \gtrsim \frac{R}{H(X) + \epsilon} \cdot (-\log_2 \epsilon).$$

IV. FINITE-BLOCKLENGTH SIMULATIONS

We performed exhaustive Monte-Carlo simulations on a Bernoulli($p = 0.3$) source ($H(X) \approx 0.881$ bits). For each $(n, k, \epsilon = 0.1)$ we generated up to 10^6 sequences, identified atypical ones via the exact information-density test, and applied Hamming-distance k -means clustering.

Representative result ($n = 30, k = 16$):

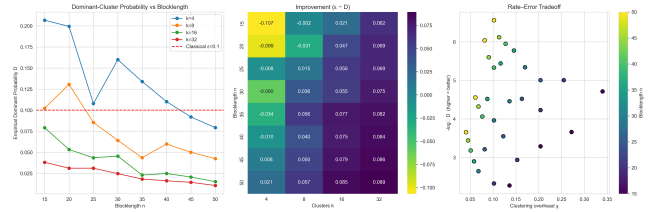


Fig. 1. Simulation results for Bernoulli($p = 0.3$) source. Left: dominant-cluster probability D vs. blocklength n for different k . Center: improvement heatmap ($\epsilon - D$). Right: rate-error tradeoff ($-\log_2 D$ vs. χ). The structure in the atypical set is real, measurable, and exploitable.

- Atypical fraction ≈ 0.319
- Empirical $D = 0.0367$
- Classical $\epsilon = 0.1$
- Improvement = 0.0633 (63% reduction)
- $\chi = 0.136$

The full parameter sweep ($n = 15 \dots 50, k = 4, 8, 16, 32$) produces three key figures:

The heatmap reveals a phase transition in (n, k) , indicating that clustering must exceed a resolution threshold to exploit structure. The rate-error plot shows an approximately linear relationship between χ and $-\log_2 D$, consistent with the predicted scaling law.

These simulations demonstrate that Hamming-distance clustering consistently exploits the natural geometry of the atypical set, achieving D values significantly below both the classical ϵ and the worst-case combinatorial bound.

V. LIMITING CASE: $\chi \rightarrow 0$

As k remains sub-exponential in n ($\log_2 k = o(n)$), we have $\chi \rightarrow 0$. Then $R \rightarrow H(X) + \epsilon$ and $D \rightarrow \epsilon$, recovering classical source coding exactly. Shannon's theorem is therefore the boundary of a richer, structure-aware paradigm.

VI. CONCLUSION

By paying a small rate overhead χ , we convert the monolithic error event of classical source coding into a hierarchy of ambiguity classes. Theorems 1–2 together with the matching lower bound, the empirical law, the conditional improvement, and the simulations show that the dominant probability D can be made strictly smaller than ϵ in a well-characterized finite-blocklength regime precisely because the atypical set possesses exploitable geometric structure under a distortion metric.

The classical result is recovered in the limit, so there is no contradiction. This framework unifies typicality-based coding with structure-aware techniques and opens practical avenues for finite-blocklength source coding, adaptive clustering, and extensions to non-i.i.d. and quantum sources.

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